Minerals and Energy Resources

Social Science — Geography • Chapter 5 • *Chapter Summary & Study Guide for Daksh*

From the iron in your bicycle to the mica in your toothpaste, from the petrol in vehicles to the electricity in your phone — almost every modern artefact contains a mineral or runs on an energy resource. This chapter explains how minerals are formed, where India's main mineral belts lie, the urgency of conservation, and the shift from conventional (coal, petroleum) to non-conventional (solar, wind, nuclear) energy sources.

Minerals — types and occurrence

A **mineral** is a homogeneous, naturally-occurring substance with a definable chemical composition. An **ore** is a mineral from which the metal can be extracted economically.

- **Metallic** — Ferrous (iron-based: iron ore, manganese, nickel, cobalt) and Non-ferrous (copper, bauxite, lead, gold).
- **Non-metallic** — mica, salt, potash, limestone, sulphur, marble, sandstone.
- **Energy minerals** — coal, petroleum, natural gas.

Minerals occur in **veins and lodes** in igneous/metamorphic rocks (tin, copper, zinc, lead), in **sedimentary beds** (coal, iron ore, gypsum, potash), through **residual decomposition** of surface rocks (bauxite), in **placer deposits** in valley sands (gold, platinum), and as ocean-bed minerals (manganese nodules, salt from sea water).

Mineral belts of India

- **North-Eastern Plateau** (Chhotanagpur — parts of Jharkhand, Odisha, Chhattisgarh, West Bengal) — iron ore, coal, manganese, bauxite, mica. India's mineral heartland.
- **South-Western Plateau** (Karnataka, Goa, parts of Tamil Nadu) — iron ore, manganese, limestone; some gold (Kolar).
- **North-Western region** (Aravalli range — Rajasthan and parts of Gujarat) — copper, zinc, lead; sandstone, granite, marble; salt from coastal Gujarat.

Important minerals

- **Iron ore** — Odisha (Mayurbhanj), Jharkhand (Singhbhum), Chhattisgarh (Bailadila), Karnataka (Bellary, Kudremukh).
- **Manganese** — used in making steel and dry batteries; Odisha (largest), MP, Karnataka.
- **Copper** — used in electrical wiring; Singhbhum (Jharkhand), Khetri (Rajasthan), Balaghat (MP).
- **Bauxite** — main ore of aluminium; Odisha (Panchpatmali), Jharkhand, MP, Gujarat.
- **Mica** — used in electronics; Jharkhand (Koderma), Andhra (Nellore), Rajasthan.
- **Limestone** — for cement and steel industries; widely distributed.

Conservation of minerals

Minerals take millions of years to form, so they are non-renewable on human time-scales. Conservation methods: improved technologies for ore extraction; use of substitutes; recycling of metals; reducing waste in mining; finding new sources/lower-grade ores.

Energy resources — conventional

- **Coal** — India's most important fuel source; provides ~55% of energy. Types: anthracite (highest carbon, rare), bituminous, lignite (Neyveli), peat. Major fields: Jharia, Bokaro, Raniganj (Jharkhand-WB), Korba (Chhattisgarh), Talcher (Odisha), Singrauli (MP).
- **Petroleum** — used as fuel and feedstock for petrochemicals. Mumbai High (offshore), Gujarat (Ankleshwar), Assam (Digboi, Naharkatiya). India imports the bulk of its petroleum.
- **Natural gas** — clean fuel, lower carbon emissions; Krishna-Godavari basin, Mumbai High, Gujarat. CNG in vehicles.

- **Hydroelectricity** — renewable; from large dams (Bhakra, Hirakud, Tehri).

Energy resources — non-conventional

- **Solar energy** — India is a tropical country with high potential. Photovoltaic panels and solar thermal plants in Rajasthan, Gujarat.
- **Wind energy** — India is among the top wind-power producers; Tamil Nadu (largest farm at Kanyakumari), Gujarat, Maharashtra.
- **Biogas** — from animal/plant waste; rural Gobar Gas plants and city sewage plants.
- **Tidal energy** — Gulf of Khambhat, Gulf of Kachchh, Sundarban delta have potential.
- **Geothermal** — Manikaran (Himachal), Puga Valley (Ladakh).
- **Nuclear** — uranium (Jharkhand, Rajasthan), thorium (Kerala, AP); plants at Tarapur, Rawatbhata, Kakrapar, Kalpakkam, Kaiga, Narora, Kudankulam.

Energy conservation — the way forward

We must use energy efficiently — public transport over private; LED bulbs; star-rated appliances; building design for natural light/ventilation; switching off when not in use; promoting renewables; carpooling. Climate change makes this urgent — fossil fuel use must drop sharply over the coming decades.

★ MUST-REMEMBER POINTS

- Mineral types: ferrous, non-ferrous metallic; non-metallic; energy minerals.
- Three mineral belts of India: NE plateau (Chhotanagpur — biggest), SW plateau, NW Aravalli.
- Iron ore: Bailadila, Bellary, Singhbhum; Bauxite: Panchpatmali; Coal: Jharia, Raniganj.
- Petroleum: Mumbai High (offshore), Ankleshwar, Digboi.
- Conventional: coal (55% energy), petroleum, natural gas, hydro.
- Non-conventional: solar, wind, biogas, tidal, geothermal, nuclear.
- Tamil Nadu has the largest wind farm (Kanyakumari area).

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